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# Integration Manual

for MPC560XB SPI Driver

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Rev 1.2





# Contents

| Section number                  | Title                                               | Page |
|---------------------------------|-----------------------------------------------------|------|
| <b>Chapter 1</b>                |                                                     |      |
| <b>Revision History</b>         |                                                     |      |
| <b>Chapter 2</b>                |                                                     |      |
| <b>Introduction</b>             |                                                     |      |
| 2.1                             | Supported Derivatives.....                          | 8    |
| 2.2                             | Overview.....                                       | 8    |
| 2.3                             | About this Manual.....                              | 8    |
| 2.4                             | Acronyms and Definitions.....                       | 9    |
| 2.5                             | Reference List.....                                 | 9    |
| <b>Chapter 3</b>                |                                                     |      |
| <b>Building the Driver</b>      |                                                     |      |
| 3.1                             | Build Options.....                                  | 11   |
| 3.1.1                           | GHS Compiler/Linker/Assembler Options.....          | 11   |
| 3.1.2                           | DIAB Compiler/Linker/Assembler Options.....         | 13   |
| 3.1.3                           | CW Compiler/Linker/Assembler Options.....           | 14   |
| 3.1.4                           | HT Compiler/Linker/Assembler Options.....           | 16   |
| 3.2                             | Files required for Compilation.....                 | 17   |
| 3.3                             | Setting up the Plug-ins.....                        | 19   |
| <b>Chapter 4</b>                |                                                     |      |
| <b>Function Calls to Module</b> |                                                     |      |
| 4.1                             | Function Calls during Start-up.....                 | 21   |
| 4.2                             | Function Calls during Shutdown.....                 | 21   |
| 4.3                             | Function Calls during Wake-up.....                  | 21   |
| <b>Chapter 5</b>                |                                                     |      |
| <b>Module Requirements</b>      |                                                     |      |
| 5.1                             | Exclusive areas to be defined in BSW scheduler..... | 23   |
| 5.2                             | SPI_EXCLUSIVE_AREA_01.....                          | 23   |
| 5.3                             | SPI_EXCLUSIVE_AREA_02.....                          | 23   |

| Section number | Title                                          | Page |
|----------------|------------------------------------------------|------|
| 5.4            | SPI_EXCLUSIVE_AREA_03.....                     | 24   |
| 5.5            | SPI_EXCLUSIVE_AREA_04.....                     | 24   |
| 5.6            | SPI_EXCLUSIVE_AREA_05.....                     | 24   |
| 5.7            | SPI_EXCLUSIVE_AREA_06.....                     | 24   |
| 5.8            | Critical region exclusivity matrix.....        | 24   |
| 5.9            | Peripheral Hardware Requirements.....          | 25   |
| 5.10           | DMA configuration.....                         | 25   |
| 5.11           | ISR to configure within OS – dependencies..... | 28   |
| 5.12           | ISR macro.....                                 | 29   |
| 5.13           | Other AUTOSAR modules - dependencies.....      | 30   |

## Chapter 6 Main API Requirements

|     |                                                           |    |
|-----|-----------------------------------------------------------|----|
| 6.1 | Main functions calls within BSW scheduler.....            | 33 |
| 6.2 | Calls to notification functions, callbacks, callouts..... | 33 |

## Chapter 7 Memory Allocation

|     |                                         |    |
|-----|-----------------------------------------|----|
| 7.1 | Sections to be defined in MemMap.h..... | 35 |
| 7.2 | Linker command file.....                | 38 |

## Chapter 8 Configuration Parameter Considerations

## Chapter 9 Integration Steps

# Chapter 1

## Revision History

**Table 1-1. Document Change History**

| Date        | Version | Changed by        | Change description           |
|-------------|---------|-------------------|------------------------------|
| 17-Feb-2012 | 1.0     | Subramanya M Naik | Initial Release              |
| 14-May-2013 | 1.1     | Manojraj L        | Update for RTM 1.0.0 release |
| 15-May-2014 | 1.2     | Hoang Ngoc Hien   | Update for RTM 1.0.1 release |



## Chapter 2

# Introduction

This Integration Manual describes the integration requirements for Autosar SPI Driver for Freescale Semiconductor's MPC560XB microcontrollers .

The roadmap for the document is as follows:

**Building the Driver** : This section gives a brief overview of the build procedure (compiler, linker options and source files) and Plugins setup.

**Function Calls to Module** : This section lists the various function calls to modules during Start-up, Shutdown and Wake-up.

**Module Requirements** : This section specifies the various module requirements related to

- Exclusive areas to be defined in BSW scheduler
- Peripheral Hardware Requirements
- Specific interface to other modules
- ISR to configure within OS
- Dependencies with other AUTOSAR modules

**Main API Requirements** : This section specifies the requirements related to the main SPI\_main API and gives a brief overview of the main functions calls within BSW scheduler, API\_Name Requirements and calls to notification functions, callbacks, callouts.

**Memory Allocation** : This section describes the memory allocation requirements namely the sections to be defined in MemMap.h and the linker command file.

**Configuration Parameter Considerations** : This section covers the various Pre Compile, Link Time and Post Build time configuration parameters.

**Integration Steps** : This section describes in brief the steps for integrating SPI module.

## 2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of Freescale Semiconductor:

**Table 2-1. Supported Derivatives**

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Freescale Semiconductor | mpc5601dxlh_lqfp64<br>mpc5601dxll_lqfp100<br>mpc5602dxlh_lqfp64<br>mpc5602dxll_lqfp100<br>mpc5602bxlh_lqfp64<br>mpc5602bxll_lqfp100<br>mpc5602bxlq_lqfp144<br>mpc5602cxlh_lqfp64<br>mpc5602cxll_lqfp100<br>mpc5603bxlh_lqfp64<br>mpc5603bxll_lqfp100<br>mpc5603bxlq_lqfp144<br>mpc5603cxlh_lqfp64<br>mpc5603cxll_lqfp100<br>mpc5604bxlh_lqfp64<br>mpc5604bxll_lqfp100<br>mpc5604bxlq_lqfp144<br>mpc5604bxmg_mapbga208<br>mpc5604cxlh_lqfp64<br>mpc5604cxll_lqfp100<br>mpc5605bxll_lqfp100<br>mpc5605bxlq_lqfp144<br>mpc5605bxlu_lqfp176<br>mpc5606bkll_lqfp100<br>mpc5606bxlq_lqfp144<br>mpc5606bxlu_lqfp176<br>mpc5607bxlu_lqfp176<br>mpc5607bxmg_mapbga208 |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

All of the above microcontroller devices are collectively named as MPC560XB.

## 2.2 Overview

AUTOSAR (Automotive Open System Architecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

## 2.3 About this Manual

This Technical Reference employs the following typographical conventions:

**Boldface type:** Bold is used for important terms, notes and warnings.

*Italic font:* Italic typeface is used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

## 2.4 Acronyms and Definitions

**Table 2-2. Acronyms and Definitions**

| Term       | Definition                        |
|------------|-----------------------------------|
| BSW        | Basic Software                    |
| DEM        | Diagnostic Event Manager          |
| DET        | Development Error Tracer          |
| ECU        | Electronic Control Unit           |
| SPI        | Serial Peripheral Interface       |
| ISR        | Interrupt Service Routine         |
| OS         | Operating System                  |
| MCU        | Microcontroller Unit              |
| GUI        | Graphical User Interface          |
| API        | Application Programming Interface |
| PB Variant | Post Build Variant                |
| PC Variant | Pre Compile Variant               |
| LT Variant | Link Time Variant                 |

## 2.5 Reference List

**Table 2-3. Reference List**

| Title                                              | Version                  |
|----------------------------------------------------|--------------------------|
| AUTOSAR SPI Driver Software Specification Document | V3.2.0                   |
| MPC5607B Reference Manual                          | Rev. 7.2, May 2012       |
| MPC5604B Reference Manual                          | Rev. 8.2, September 2013 |
| 5606B Reference Manual                             | Rev. 2, May 2014         |
| 5602D Reference Manual                             | Rev. 4.2, Dec 2013       |
| Bolero 1.5M cut 2.2 Errata Sheet                   | Rev. 12Dec2013           |

*Table continues on the next page...*

**Table 2-3. Reference List (continued)**

| Title                                                | Version            |
|------------------------------------------------------|--------------------|
| (MPC5607B_1M03Y)                                     |                    |
| Bolero 512k cut 2.3 Errata Sheet<br>(MPC5604B_2M27V) | Rev. 26 FEB 2014   |
| Bolero 1M cut 1.0 Errata Sheet<br>(MPC5606B_0N13E)   | Rev. , 12 DEC 2013 |
| Bolero 256k cut 1.1 Errata Sheet<br>(MPC5602D_1M18Y) | Rev. , 12 DEC 2013 |

## Chapter 3

# Building the Driver

This section describes the source files and various compiler, linker options used for building the Autosar SPIDriver for Freescale Semiconductor's MPC560XB microcontrollers. It also explains the Plugins setup procedure.

### NOTE

The Spi\_TS\_T2D13M10I1R0 is composed as follow:  
TS\_T<Target\_Id>D<Derivative\_Id>M<SW\_Version\_Major>I  
<SW\_Version\_Minor>R0 (i.e. Target\_Id = 2 identifies  
PowerPC architecture and Derivative\_Id = 13 identifies the  
MPC560XB

## 3.1 Build Options

The driver files are compiled using Green Hills MULTI v6.1.4 , Windriver DIAB Rel DIAB\_5\_9\_2\_0-FCS\_20120725\_234511, Freescale CodeWarrior for MPC5xxx v2.10 - compiler Version 4.3 build 224, HighTec v4.6.4.0, Compiler gcc v4.6.3 The compiler, linker flags used for building the driver are explained below:

### 3.1.1 GHS Compiler/Linker/Assembler Options

Table 3-1. Compiler Options

| Option        | Description                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------|
| -cpu=ppc560xb | Selects target processor: ppc560xb                                                                                               |
| -ansi         | Specifies ANSI C with extensions. This mode extends the ANSI X3.159-1989 standard with certain useful and compatible constructs. |
| -noSPE        | Disables the use of SPE and vector floating point instructions by the compiler.                                                  |
| -Ospace       | Optimize for size                                                                                                                |
| -sda=0        | Enables the Small Data Area optimization with a threshold of 0.                                                                  |
| -vle          | Enables VLE code generation                                                                                                      |

*Table continues on the next page...*

**Table 3-1. Compiler Options (continued)**

| Option                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -dual_debug                          | Enables the generation of DWARF, COFF, or BSD debugging information in the object file                                                                                                                                                                                                                                                                                                                                            |
| -G                                   | Generates source level debugging information and allows procedure call from debugger's command line.                                                                                                                                                                                                                                                                                                                              |
| --no_exceptions                      | Disables support for exception handling                                                                                                                                                                                                                                                                                                                                                                                           |
| -Wundef                              | Generates warnings for undefined symbols in preprocessor expressions                                                                                                                                                                                                                                                                                                                                                              |
| -Wimplicit-int                       | Issues a warning if the return type of a function is not declared before it is called                                                                                                                                                                                                                                                                                                                                             |
| -Wshadow                             | Issues a warning if the declaration of a local variable shadows the declaration of a variable of the same name declared at the global scope, or at an outer scope                                                                                                                                                                                                                                                                 |
| -Wtrigraphs                          | Issues a warning for any use of trigraphs                                                                                                                                                                                                                                                                                                                                                                                         |
| --prototype_errors                   | Generates errors when functions referenced or called have no prototype                                                                                                                                                                                                                                                                                                                                                            |
| --incorrect_pragma_warnings          | Valid #pragma directives with wrong syntax are treated as warnings                                                                                                                                                                                                                                                                                                                                                                |
| -noslashcomment                      | C++ like comments will generate a compilation error                                                                                                                                                                                                                                                                                                                                                                               |
| -preprocess_assembly_files           | Preprocesses assembly files                                                                                                                                                                                                                                                                                                                                                                                                       |
| --short_enum                         | Store enumerations in the smallest possible type                                                                                                                                                                                                                                                                                                                                                                                  |
| --no_commons                         | Allocates uninitialized global variables to a section and initializes them to zero at program startup.                                                                                                                                                                                                                                                                                                                            |
| -c                                   | Produces an object file (called input-file.o) for each source file.                                                                                                                                                                                                                                                                                                                                                               |
| -DAUTOSAR_OS_NOT_USED                | -D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options                                                                                                                          |
| -DUSE_SW_VECTOR_MODE                 | -D defines a preprocessor symbol and optionally can set it to a value. USE_SW_VECTOR_MODE: By default in the package, drivers are compiled to be used with interrupt controller configured to be in hardware vector mode. In case of AUTOSAR_OS_NOT_USED, the compiler option "-DUSE_SW_VECTOR_MODE" must be added to the list of compiler options to be used with interrupt controller configured to be in software vector mode. |
| -DDISABLE_MCAL_INTERMODULE_ASR_CHECK | -D defines a preprocessor symbol to disable the inter-module version check for AR_RELEASE versions. DISABLE_MCAL_INTERMODULE_ASR_CHECK: By default in the package, drivers are compiled to perform the inter-module version check as per Autosar BSW004. When the inter-module version check needs to be disabled then the DISABLE_MCAL_INTERMODULE_ASR_CHECK global define must be added to the list of compiler options.        |
| -DGHS                                | -D defines a preprocessor symbol and optionally can set it to a value. This one defines the GHS preprocessor symbol.                                                                                                                                                                                                                                                                                                              |

**Table 3-2. Assembler Options**

| Option        | Description                        |
|---------------|------------------------------------|
| -cpu=ppc560xb | Selects target processor: ppc560xb |

**Table 3-3. Linker Options**

| Option           | Description                        |
|------------------|------------------------------------|
| -cpu=ppc560xb    | Selects target processor: ppc560xb |
| -nostartfiles    | Do not use Start files.            |
| -vle             | Enables VLE code generation        |
| -linker_warnings | Display linker warnings            |

### 3.1.2 DIAB Compiler/Linker/Assembler Options

**Table 3-4. Compiler Options**

| Option                 | Description                                                                                                                                                                                                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -Xdialect-ansi         | Follow the ANSI C standard with some additions                                                                                                                                                                                                                                                           |
| -XO                    | Enables extra optimizations to produce highly optimized code                                                                                                                                                                                                                                             |
| -g                     | Generate symbolic debugger information and disable some optimizations.                                                                                                                                                                                                                                   |
| -Xsize-opt             | Optimize for size rather than speed when there is a choice                                                                                                                                                                                                                                               |
| -Xsmall-data=0         | Set Size Limit for "small data" Variables to zero.                                                                                                                                                                                                                                                       |
| -Xaddr-sconst=0x11     | Specify addressing for constant static and global variables with size less than or equal to -Xsmall-const to far-absolute.                                                                                                                                                                               |
| -Xaddr-sdata=0x11      | Specify addressing for non-constant static and global variables with size less than or equal to -Xsmall-data in size to far-absolute.                                                                                                                                                                    |
| -Xno-common            | Disable use of the "COMMON" feature so that the compiler or assembler will allocate each uninitialized public variable in the .bss section for the module defining it, and the linker will require exactly one definition of each public variable                                                        |
| -Xnested-interrupts    | Allow nested interrupts                                                                                                                                                                                                                                                                                  |
| -Xdebug-dwarf2         | Generate symbolic debug information in dwarf2 format                                                                                                                                                                                                                                                     |
| -Xdebug-local-all      | Force generation of type information for all local variables                                                                                                                                                                                                                                             |
| -Xdebug-local-cie      | Create common information entry per module                                                                                                                                                                                                                                                               |
| -Xdebug-struct-all     | Force generation of type information for all typedefs, struct, union and class types                                                                                                                                                                                                                     |
| -Xforce-declarations   | Generates warnings if a function is used without a previous declaration                                                                                                                                                                                                                                  |
| -ee1481                | Generate an error when the function was used before it has been declared                                                                                                                                                                                                                                 |
| -Xmacro-undefined-warn | Generates a warning when an undefined macro name occurs in a #if preprocessor directive                                                                                                                                                                                                                  |
| -W:as;,-l              | Pass the option "-l" (lower case letter L) to the assembler to get an assembler listing file                                                                                                                                                                                                             |
| -Wa,-Xisa-vle          | Instruct the assembler to expect and assemble VLE (Variable Length Encoding) instructions rather than BookE instructions.                                                                                                                                                                                |
| -DAUTOSAR_OS_NOT_USED  | -D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options |
| -DDIAB                 | -D defines a preprocessor symbol and optionally can set it to a value. This one defines the DIAB preprocessor symbol.                                                                                                                                                                                    |
| -Xforce-prototypes     | -Generate warnings if a function is used without a previous prototype declaration.                                                                                                                                                                                                                       |

*Table continues on the next page...*

**Table 3-4. Compiler Options (continued)**

| Option                | Description                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -tPPCE200Z0VEN:simple | Sets target processor to PPCE200Z0, generates ELF using EABI conventions, No floating point support (minimizes the required runtime), selects simple environment settings for Startup Module and Libraries. |

**Table 3-5. Assembler Options**

| Option                | Description                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -tPPCE200Z0VEN:simple | Sets target processor to PPCE200Z0, generates ELF using EABI conventions, No floating point support (minimizes the required runtime), selects simple environment settings for Startup Module and Libraries                                                                                                                                             |
| -g                    | Dump the symbols in the global symbol table in each archive file.                                                                                                                                                                                                                                                                                      |
| -Xisa-vle             | Expect and assemble VLE (Variable Length Encoding) instructions rather than Book E instructions. The default code section is named .text_vle instead of .text, and the default code section fill "character" is set to 0x44444444 instead of 0. The .text_vle code section will have ELF section header flags marking it as VLE code, not Book E code. |

**Table 3-6. Linker Options**

| Option                | Description                                                                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -tPPCE200Z0VEN:simple | Sets target processor to PPCE200Z0, generates ELF using EABI conventions, No floating point support (minimizes the required runtime), selects simple environment settings for Startup Module and Libraries |
| -Xelf                 | Generates ELF object format for output file                                                                                                                                                                |
| -m6                   | Generates a detailed link map and cross reference table                                                                                                                                                    |
| -lc                   | Specifies to linker to search for libc.a                                                                                                                                                                   |
| -Xlibc-old            | Enables usage of legacy (pre-release 5.6) libraries                                                                                                                                                        |

### 3.1.3 CW Compiler/Linker/Assembler Options

**Table 3-7. Compiler Options**

| Option            | Description                                                                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -proc Zen         | Generates and links object code for Zen processor. The compiler uses unsigned as the default parameter for the -char switch                                                                                                     |
| -lang c           | Expects source code to conform to the language specified by the ISO/IEC 9899-1990 ("C90") standard                                                                                                                              |
| -opt all          | This option is selected all optimization (the same as -opt speed,level=4,intrinsics,noframe)                                                                                                                                    |
| -common off       | Disables moving uninitialized data into a common section                                                                                                                                                                        |
| -sdatathreshold 0 | Specifies the threshold size (in bytes) for an item considered by the linker to be small data. (The linker stores small data items in the Small Data address space. The compiler can generate faster code to access this data.) |

*Table continues on the next page...*

**Table 3-7. Compiler Options (continued)**

| Option                | Description                                                                                                                                                                                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -sdata2threshold 0    | Specifies the threshold size (in bytes) for an item considered by the linker to be small constant data. (The linker stores small constant data items in the Small Constant Data address space.)                                                                                                          |
| -vle                  | Tells the compiler and linker to generate and lay out Variable Length Encoded (VLE) instructions, available on Zen variants of Power Architecture processors                                                                                                                                             |
| -use_lmw_stmw on      | Enables the use of multiple load and store instructions for function prologues and epilogues                                                                                                                                                                                                             |
| -ppc_asm_to_vle       | Converts regular Power Architecture assembler mnemonics to equivalent VLE (Variable Length Encoded) assembler mnemonics in the inline assembler                                                                                                                                                          |
| -Cpp_exceptions off   | When on, generates executable code for C++ exceptions. When off, generates smaller, faster executable code                                                                                                                                                                                               |
| -func_align 4         | Specifies alignment of functions in executable code                                                                                                                                                                                                                                                      |
| -sym dwarf-2,full     | Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format)                                                                                                                                                                                                                   |
| -gdwarf-2             | Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format). The linker ignores debugging information that is not in the Dwarf 1, Dwarf 2 format                                                                                                                              |
| -w on                 | Turns on most warning messages                                                                                                                                                                                                                                                                           |
| -r                    | Compiler should expect function prototypes                                                                                                                                                                                                                                                               |
| -w undefmacro         | Issues warning messages on the use of undefined macros in #if and #elif conditionals                                                                                                                                                                                                                     |
| -char unsigned        | Controls the default sign of the char data type: char data items are unsigned                                                                                                                                                                                                                            |
| -nosyspath            | Performs a search of both the user and system paths, treating #include statements of the form #include xyz the same as the form #include "xyz"                                                                                                                                                           |
| -fp none              | No floating point code generation                                                                                                                                                                                                                                                                        |
| -DAUTOSAR_OS_NOT_USED | -D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options |
| -DMWERKS              | -D defines a preprocessor symbol and optionally can set it to a value. This one defines the CW preprocessor symbol.                                                                                                                                                                                      |

**Table 3-8. Assembler Options**

| Option            | Description                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -proc Zen         | Generates and links object code for Zen processor. The compiler uses unsigned as the default parameter for the -char switch                                                  |
| -vle              | Tells the compiler and linker to generate and lay out Variable Length Encoded (VLE) instructions, available on Zen variants of Power Architecture processors                 |
| -sym dwarf-2,full | Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format)                                                                                       |
| -gdwarf-2         | Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format). The linker ignores debugging information that is not in the Dwarf 1, Dwarf 2 format. |
| -DMWERKS          | -D defines a preprocessor symbol and optionally can set it to a value. This one defines the CW preprocessor symbol.                                                          |

**Table 3-9. Linker Options**

| Option               | Description                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| -proc Zen            | Generates and links object code for Zen processor. The compiler uses unsigned as the default parameter for the -char switch |
| -code_merging all    | Removes duplicated functions to reduce object code size                                                                     |
| -far_near_addressing | Simplifies address computations to reduce object code size and improve performance                                          |
| -vle_enhance_merging | Removes duplicated functions that are called by functions that use VLE instructions to reduce object code size              |
| -listdwarf           | DWARF debugging information in the linker's map file                                                                        |
| -sym dwarf-2,full    | Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format)                                      |
| -char unsigned       | Controls the default sign of the char data type: char data items are unsigned.                                              |

### 3.1.4 HT Compiler/Linker/Assembler Options

**Table 3-10. Compiler Options**

| Option                | Description                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -Os                   | Optimize for size. '-Os' enables all '-O2' optimizations that do not typically increase code size. It also performs further optimizations designed to reduce code size.                                                                                                                                                                                                                             |
| -msdata=eabi          | On System V.4 and embedded PowerPC systems, put small initialized const global and static data in the '.sdata2' section, which is pointed to by register r2. Put small initialized non-const global and static data in the '.sdata' section, which is pointed to by register r13. Put small uninitialized global and static data in the '.sbss' section, which is adjacent to the '.sdata' section. |
| -G1000                | On embedded PowerPC systems, put global and static items less than or equal to num bytes into the small data or bss sections instead of the normal data or bss section.                                                                                                                                                                                                                             |
| -fshort-double        | Use the same size for double as for float.                                                                                                                                                                                                                                                                                                                                                          |
| -mno-spe              | This option used to disable spe extension (Like -noSPE on ghs compiler).                                                                                                                                                                                                                                                                                                                            |
| -mregnames            | On System V.4 and embedded PowerPC systems do (do not) emit register names in the assembly language output using symbolic forms                                                                                                                                                                                                                                                                     |
| -g3                   | Request debugging information of Level 3. Includes extra information, such as all the macro definitions present in the program.                                                                                                                                                                                                                                                                     |
| -Wall                 | This enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros.                                                                                                                                                                                                            |
| -mvle                 | Enable support of VLE                                                                                                                                                                                                                                                                                                                                                                               |
| --mcpu=z425n3         | Support of e200 Z425n3 core                                                                                                                                                                                                                                                                                                                                                                         |
| -DAUTOSAR_OS_NOT_USED | -D defines a preprocessor symbol and optionally can set it to a value.<br>AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options                                                                                         |
| -DUSE_SW_VECTOR_MODE  | -D defines a preprocessor symbol and optionally can set it to a value.<br>USE_SW_VECTOR_MODE: By default in the package, drivers are compiled to be used with interrupt controller configured to be in hardware vector mode. In case of AUTOSAR_OS_NOT_USED, the compiler option "-DUSE_SW_VECTOR_MODE" must be                                                                                     |

**Table 3-10. Compiler Options**

| Option | Description                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------------------|
|        | added to the list of compiler options to be used with interrupt controller configured to be in software vector mode. |

**Table 3-11. Assembler Options**

| Option        | Description                                                                                                                                                                                                                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -c            | Compile or assemble the source files, but do not link. The linking stage simply is not done. The ultimate output is in the form of an object file for each source file.                                                                                                                                                                         |
| -Wa,--gdwarf2 | Produce debugging information in DWARF format.                                                                                                                                                                                                                                                                                                  |
| -msdata=eabi  | Put small initialized const global and static data in the .sdata2 section, which is pointed to by register r2. Put small initialized non-const global and static data in the .sdata section, which is pointed to by register r13. Put small uninitialized global and static data in the .sbss section, which is adjacent to the .sdata section. |
| -mregnames    | Do emit register names in the assembly language output using symbolic forms.                                                                                                                                                                                                                                                                    |
| -mno-spe      | This option used to disable spe extension (Like --noSPE on ghs compiler).                                                                                                                                                                                                                                                                       |
| -mcpu=z425n3  | Support of e200 z425n3 core                                                                                                                                                                                                                                                                                                                     |

**Table 3-12. Linker Options**

| Option                     | Description                                                                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -N                         | Set the text and data sections to be readable and writable. Also, do not page-align the data segment. If the output format supports Unix style magic numbers, mark the output as OMAGIC.                                                                 |
| --no-undefined             | Normally when a symbol has an undefined version, the linker will ignore it. This option disallows symbols with undefined version and a fatal error will be issued instead.                                                                               |
| --warn-common              | Normally ppc-vle-ld will give a warning if it finds an incompatible library during a library search. This option silences the warning.                                                                                                                   |
| --error-unresolved-symbols | This restores the linker's default behaviour of generating errors when it is reporting unresolved symbols.                                                                                                                                               |
| -nocrt0                    | library crt0 from compiler library not used, it uses custom crt0 file                                                                                                                                                                                    |
| -nostartfiles              | no start file from compiler library is used                                                                                                                                                                                                              |
| -Wl                        | If the linker is being invoked indirectly, via a compiler driver (e.g. 'gcc') then all the linker command line options should be prefixed by '-Wl,'Used in combination with -no-warn-flags, --no-undefined, --warn-common and --error-unresolved-symbols |
| --no-warn-flags            | Used in combination with -Wl disable linker warnings. See previous                                                                                                                                                                                       |
| -T                         | The -T option specifies the name of your own script ie sprig.link. The -c is a synonym of the -T option                                                                                                                                                  |

## 3.2 Files required for Compilation

This section describes the include files required to compile, assemble (if assembler code) and link the Autosar SPI driver for Freescale Semiconductor's MPC560XB microcontrollers.

To avoid integration of incompatible files, all the include files from other modules shall have the same AR\_MAJOR\_VERSION and AR\_MINOR\_VERSION, i.e. only files with the same Autosar major and minor versions can be compiled.

### SPI Files:

```
..\Spi_TS_T2D13M10I1R0\include\Spi.h
..\Spi_TS_T2D13M10I1R0\include\Spi_LLD.h
..\Spi_TS_T2D13M10I1R0\include\Spi_Dspi_LLD.h
..\Spi_TS_T2D13M10I1R0\include\Spi_Dspi_LLD_CfgEx.h
..\Spi_TS_T2D13M10I1R0\include\Dma_LLD.h
..\Spi_TS_T2D13M10I1R0\include\Reg_eSys_DSPI.h
..\Spi_TS_T2D13M10I1R0\include\Reg_eSys_DMA.h

..\Spi_TS_T2D13M10I1R0\src\Spi.c
..\Spi_TS_T2D13M10I1R0\src\Spi_Irq.c
..\Spi_TS_T2D13M10I1R0\src\Spi_Dspi_LLD.c
..\Spi_TS_T2D13M10I1R0\src\Spi_Dma_LLD.c
```

Spi\_Cfg.c (For PC Variant) - This file should be generated by the user using a configuration tool for compilation

Spi\_Lcfg.c (For LT Variant) - This file should be generated by the user using a configuration tool for compilation

Spi\_PBcfg.c (For PB Variant) - This file should be generated by the user using a configuration tool for compilation

Spi\_Cfg.h - This file should be generated by the user using a configuration tool for compilation

### Other include files:

### Files from common folder:

```
..\Base_TS_T2D13M10I1R0\include\*.h
..\Base_TS_T2D13M10I1R0\generate_PC\include\modules.h
..\Base_TS_T2D13M10I1R0\generate_PC\include\Soc_Ips.h
```

**Files from Dem folder:**

```
..\Dem_TS_T2D13M10I1R0\include\Dem.h
..\Dem_TS_T2D13M10I1R0\include\Dem_Types.h
..\Dem_TS_T2D13M10I1R0\include\generate_PC\Dem_IntErrId.h
..\Dem_TS_T2D13M10I1R0\src\Dem.c
```

**Files from Det folder:**

```
..\Det_TS_T2D13M10I1R0\include\Det.h
..\Det_TS_T2D13M10I1R0\src\Det.c
```

**Files from Rte folder(Only when OS is used):**

```
..\SchM_TS_T2D13M10I1R0\include\SchM_Spi.h
..\SchM_TS_T2D13M10I1R0\src\SchM_Spi.c
```

### 3.3 Setting up the Plug-ins

All the Autosar MCAL drivers for MPC560XB were designed to be configured using Tresos® Studio configuration and code generation tool from tresos Elektrobit Tresos Release 14.2.1 (b140128-1223).

Location of various files inside the plugin folder is explained below.

**Module Parameter Definition File:**

```
..\Spi_TS_T2D13M10I1R0\config\Spi.xdm
```

**Code Generation Templates for Pre-Compile time configuration parameters:**

```
..\Spi_TS_T2D13M10I1R0\generate_PC\src\Spi_Cfg.c
..\Spi_TS_T2D13M10I1R0\generate_PC\include\Spi_Cfg.h
```

**Code Generation Templates for Post-Build time configuration parameters:**

```
..\Spi_TS_T2D13M10I1R0\generate_PB\src\Spi_PBcfg.c
```

**Code Generation Templates for Link time configuration parameters:**

```
..\Spi_TS_T2D13M10I1R0\generate_LT\src\Spi_Lcfg.c
```

**Steps to generate configurations:**

1. Copy the module folders Resource<plugin\_name>, Spi\_TS\_T2D13M10I1R0 into the Tresos plug-in folder.
2. Set the desired Tresos Output location folder for the generated sources and header files.
3. Use the Tresos GUI to modify configuration parameters values.
4. Generate the Pre-Compile, Link time and Post-Build files.

## **Chapter 4**

# **Function Calls to Module**

### **4.1 Function Calls during Start-up**

The SPI Handler & Driver shall be initialized before the SPI peripherals are used. The API to be called for this purpose is Spi\_Init(). The MCU and PORT modules shall be initialized before SPI is initialized.

### **4.2 Function Calls during Shutdown**

SPI can be silenced by calling Spi\_DeInit().

### **4.3 Function Calls during Wake-up**

N/A



## Chapter 5

# Module Requirements

### 5.1 Exclusive areas to be defined in BSW scheduler

In the current implementation, SPI is using the services of Schedule Manager (SchM) for entering and exiting the critical regions.

SchM implementation is done by the integrators of the MCAL using OS or non-OS services.

For testing the SPI, stubs are used for SchM.

Some SPI driver global variables updates are performed by ISRs before calling the user notification functions. In order to avoid the scenario where an executing SPI driver function is preempted by an SPI ISR, which is modifying some of the global variables, some exclusive areas are defined

The ISR critical regions must not block the other critical regions to avoid deadlocks. This is ensured by exiting the ISR critical region before calling the user notification functions.

The following critical regions are used in the SPI driver:

### 5.2 SPI\_EXCLUSIVE\_AREA\_01

Used in function Spi\_SyncTransmit, to protect the status of the given sequence result. Also it protects the global variable which contains the status of the Spi\_SyncTransmit service. As stated by the Autosar, this service cannot be called when another sequence is during transmission, using this service.

### **5.3 SPI\_EXCLUSIVE\_AREA\_02**

Used in function Spi\_SyncTransmit, to protect the status of the given sequence result. Also it protects the global variable which contains the status of the Spi\_SyncTransmit service. As stated by the Autosar, this service cannot be called when another sequence is during transmission, using this service.

### **5.4 SPI\_EXCLUSIVE\_AREA\_03**

Used in the internal function Spi\_ScheduleJob, protects the schedule mechanism for the situation when a scheduling operation determined by a pending Spi\_AsyncTransmit() call may be preempted by a job scheduling requested by an ISR event. It also protect concurrent Spi\_AsyncTransmit() calls to schedule in the same time different jobs on the same DSPI unit.

### **5.5 SPI\_EXCLUSIVE\_AREA\_04**

Used in the internal function Spi\_ScheduleNextJob, protects the schedule mechanism for the situation when a scheduling operation determined by a pending Spi\_AsyncTransmit() call may be preempted by a job scheduling requested by an ISR event.

### **5.6 SPI\_EXCLUSIVE\_AREA\_05**

Used in the internal function Spi\_LockJobs, guaranties the atomicity of locking for the entire set of jobs belonging to an asynchronous sequence.

### **5.7 SPI\_EXCLUSIVE\_AREA\_06**

Used in the internal function Spi\_UnlockRemainingJobs, guaranties the atomicity of unlocking for the entire set of jobs belonging to an asynchronous sequence.

## 5.8 Critical region exclusivity matrix

Below is the table depicting the exclusivity between different critical region IDs from the SPI driver. If there is an “X” in a table, it means that those 2 critical regions cannot interrupt each other. The critical regions from interrupts are grouped in “ISR Critical Regions”. If an exclusive area is “exclusive” with the composed “ISR Critical Regions” group, it means that it is exclusive with each one of the ISR critical regions.

**Table 5-1. Critical region exclusivity matrix**

|                      | SPI_EA_01 | SPI_EA_02 | SPI_EA_03 | SPI_EA_04 | SPI_EA_05 | SPI_EA_06 | ISR Critical Regions |
|----------------------|-----------|-----------|-----------|-----------|-----------|-----------|----------------------|
| SPI_EA_01            | X         | X         |           |           |           |           |                      |
| SPI_EA_02            | X         | X         |           |           |           |           |                      |
| SPI_EA_03            |           |           | X         | X         |           |           | X                    |
| SPI_EA_04            |           |           | X         | X         |           |           | X                    |
| SPI_EA_05            |           |           |           |           | X         | X         |                      |
| SPI_EA_06            |           |           |           |           | X         | X         |                      |
| ISR Critical Regions |           |           | X         | X         |           |           | X                    |

### Note

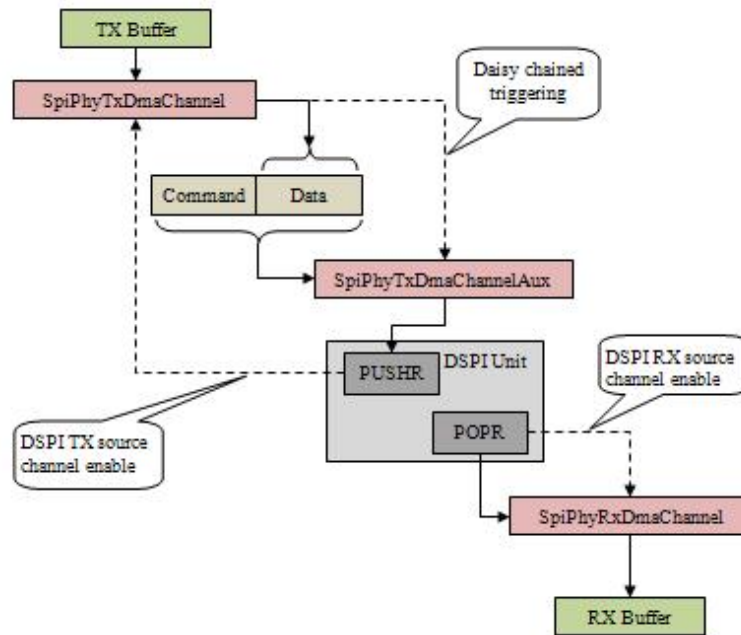
- SPI\_EA\_xx means SPI\_EXCLUSIVE\_AREA\_xx

## 5.9 Peripheral Hardware Requirements

N/A

## 5.10 DMA configuration

This section applies only to DSPI units configured for asynchronous transmission (SpiPhyUnitSync not checked) and which use DMA for serializing/deserializing data between the hardware unit and the TX/RX buffers (SpiPhyUnitAsyncMethod = DMA).



**Figure 5-1. DMA transferring mode internal architecture**

Each DSPI unit configured in DMA mode requires 3 distinct DMA channels:

**-SpiPhyTxDmaChannel:** the master TX DMA channel used for reading data from the TX buffer, in order to prepare it for serializing into an internal dataframe; this channel is triggered by TX DSPI unit event and must be “wired” to “DSPI TX source” (configured inside the MCU module – McuDMA folder) – it must be a channel linkable to the external DMA TX source for the given DSPI unit.

**-SpiPhyTxDmaChannelAux:** secondary TX DMA channel used for sending internal prepared dataframes to the DSPI unit PUSH register. This channel transfers are triggered by SpiPhyTxDmaChannel acting as master DMA channel; no specific settings needed for it – this channel must not be linked to any peripheral DMA source and not be used in any other DMA transfer.

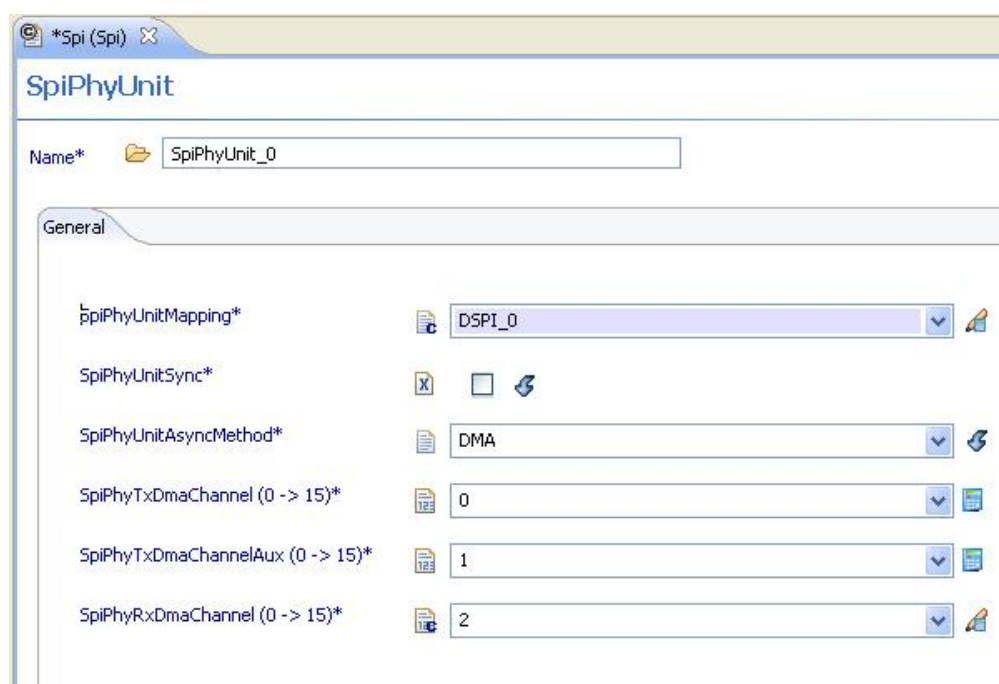
**-SpiPhyRxDmaChannel:** the RX DMA channel used for filling RX buffer with the deserialized data; this channel is triggered by RX DSPI unit event and must be “wired” to “DSPI RX source” (configured inside the MCU module – McuDMA folder) – it must be a channel linkable to the external DMA RX source for the given DSPI unit.

### Note

- If DMA uses fixed priority arbitration, then **SpiPhyTxDmaChannelAux** priority must be greater than **SpiPhyTxDmaChannel** priority.
- If DMA uses round robin arbitration, no priority constraints are applied on **SpiPhyTxDmaChannelAux** and **SpiPhyTxDmaChannel** priority.

If the SPI driver is working in interrupt mode, the DMA interrupt must be enabled for the interrupt vector corresponding to the RX DMA channel, as in section 4.5

Next figures show an example of DMA configuration for DSPI0 unit.



**Figure 5-2. DMA Configuration sample for DSPI0 Physical Unit - SPI module in tresos**

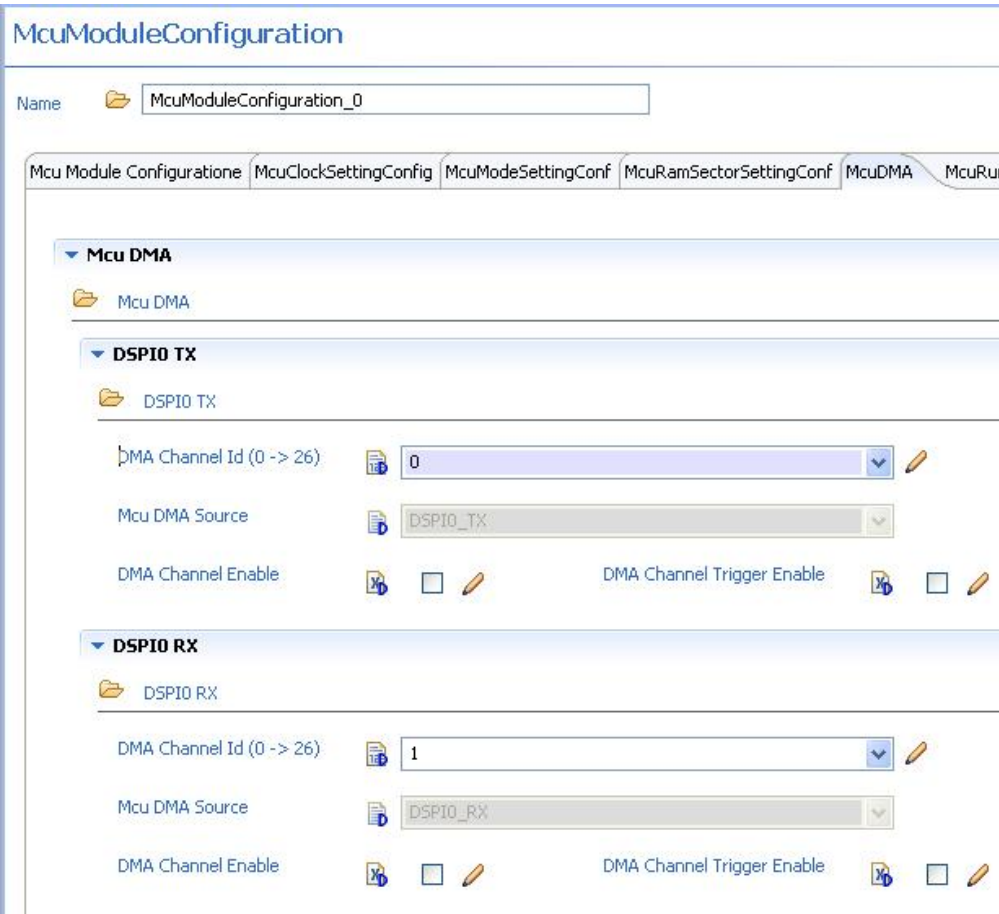


Figure 5-3. DMA channels enable sample for DSPI0 Physical Unit - MCU module in tresos

5.11 ISR to configure within OS – dependencies

The following ISRs are used by the SPI driver and need to be assigned to a priority level. The interrupt vector numbers corresponding to the DMA channel configuration is as shown in Table Table 5-2 and interrupt vector numbers corresponding to PIO\_FIFO is as shown in Table Table 5-3. The interrupt occurs each time the EOQ bit in SR register arises.

(Note: Unused interrupts shouldn't be configured in the OS.)

Table Table 5-2 shows an example Dma Vector location for the corresponding channels configured. :

Table 5-2. SPI ISRs for DMA channel

| DMA Name | DMA Interrupt Vector | DMA Channel |
|----------|----------------------|-------------|
|----------|----------------------|-------------|

Table continues on the next page...

**Table 5-2. SPI ISRs for DMA channel (continued)**

|                         |    |    |
|-------------------------|----|----|
| Spi_LLD_IsrRxDma_DSPI_0 | 13 | 2  |
| Spi_LLD_IsrRxDma_DSPI_1 | 16 | 5  |
| Spi_LLD_IsrRxDma_DSPI_2 | 19 | 8  |
| Spi_LLD_IsrRxDma_DSPI_3 | 22 | 11 |
| Spi_LLD_IsrRxDma_DSPI_4 | 25 | 14 |

**Table 5-3. SPI ISRs for PIO\_FIFO**

| ISR Name              | Hardware interrupt vector |
|-----------------------|---------------------------|
| Spi_LLD_IsrEOQ_DSPI_0 | 75                        |
| Spi_LLD_IsrEOQ_DSPI_1 | 95                        |
| Spi_LLD_IsrEOQ_DSPI_2 | 115                       |
| Spi_LLD_IsrEOQ_DSPI_3 | 183                       |
| Spi_LLD_IsrEOQ_DSPI_4 | 212                       |
| Spi_LLD_IsrEOQ_DSPI_5 | 220                       |

**Note:**In case of AUTOSAR\_OS\_NOT\_USED, the compiler option "-DUSE\_SW\_VECTOR\_MODE" must be added to the list of compiler options to be used with interrupt controller configured to be in software vector mode.

## 5.12 ISR macro

MCAL drivers use the ISR macro to define the functions that will process hardware interrupts. Depending on whether the OS is used or not, this macro can have different definitions:

1. OS is not used - AUTOSAR\_OS\_NOT\_USED is defined:

- If USE\_SW\_VECTOR\_MODE is defined:

```
#define ISR(IsrName) void IsrName(void)
```

In this case, drivers' interrupt handlers are normal C functions and the prolog/epilog handle the context save and restore.

- If USE\_SW\_VECTOR\_MODE is not defined:

```
#define ISR(IsrName) INTERRUPT_FUNC void IsrName(void)
```

In this case, drivers' interrupt handlers must save and restore the execution context.

2. Freescale Semiconductor OS is used – AUTOSAR\_OS\_NOT\_USED is not defined

```
#define ISR(IsrName) void OS_isr_##IsrName()
```

In this case, OS is handling the execution context when an interrupt occurs. Drivers' interrupt handlers are normal C functions.

3. Other vendor's OS is used – AUTOSAR\_OS\_NOT\_USED is not defined. Please refer to the OS documentation for description of the ISR macro.

Please refer to the OS documentation for description of the ISR macro.

## 5.13 Other AUTOSAR modules - dependencies

### Development Error Tracer:

This module is necessary for enabling Development error detection. The API function used is Det\_ReportError(). The activation / deactivation of Development error detection is configurable using the

'SpiDevErrorDetect' configuration parameter.

### Diagnostic Event Manager:

This module is necessary for enabling Production error detection. The API function used is Dem\_ReportErrorStatus ().

### Mcu:

The SPI reference clock is provided by MCU plugin. The reference is specified by the parameter SpiGeneral\SpiClockRef:



**Figure 5-4. Spi reference clock provided by MCU plugin**

For each DSPI in use, a transmit and a receive DMA channel need to be defined and routed through the DMA Multiplexer.

The Table [Table 5-4](#) shows an example DMA configuration. For more information, refer to section [DMA configuration](#)

**Table 5-4. SPI DMA Channel Multiplexer**

| DMA Name            | DMA Source                    |
|---------------------|-------------------------------|
| DSPI 0 Transmit DMA | 0 (DSPI0.SpiPhyTxDmaChannel)  |
| DSPI 0 Receive DMA  | 2 (DSPI0.SpiPhyRxDmaChannel)  |
| DSPI 1 Transmit DMA | 3 (DSPI1.SpiPhyTxDmaChannel)  |
| DSPI 1 Receive DMA  | 5 (DSPI1.SpiPhyRxDmaChannel)  |
| DSPI 2 Transmit DMA | 6 (DSPI2.SpiPhyTxDmaChannel)  |
| DSPI 2 Receive DMA  | 8 (DSPI2.SpiPhyRxDmaChannel)  |
| DSPI 3 Transmit DMA | 9 (DSPI3.SpiPhyTxDmaChannel)  |
| DSPI 3 Receive DMA  | 11 (DSPI3.SpiPhyRxDmaChannel) |
| DSPI 4 Transmit DMA | 12 (DSPI4.SpiPhyTxDmaChannel) |
| DSPI 4 Receive DMA  | 14 (DSPI4.SpiPhyRxDmaChannel) |

**PORT module:** For each DSPI, the SCK, SOUT, SIN and CSx\_y signals need to be configured. The following table shows an example configuration for DSPI 0 on PORT A and Port B :

**Table 5-5. SPI Pins for 5604B**

| Signal | PortPin Direction | Port Control Register (PCR) | PortPinLevelValue | PortPinMux |
|--------|-------------------|-----------------------------|-------------------|------------|
| CS1_0  | Out               | 28                          | High              | Option3    |
| CS0_0  | Out               | 15                          | High              | Option1    |
| SCK    | Out               | 14                          | High              | Option1    |
| SOUT   | Out               | 13                          | High              | Option1    |
| SIN    | In                | 12                          | High              | -          |

**Table 5-6. SPI Pins for 5607B**

| Signal | PortPin Direction | Port Control Register (PCR) | PortPinLevelValue | PortPinMux |
|--------|-------------------|-----------------------------|-------------------|------------|
| CS1_0  | Out               | 90                          | High              | Option3    |
| CS0_0  | Out               | 15                          | High              | Option1    |
| SCK    | Out               | 14                          | High              | Option1    |
| SOUT   | Out               | 13                          | High              | Option1    |
| SIN    | In                | 12                          | High              | -          |



## Chapter 6

# Main API Requirements

### 6.1 Main functions calls within BSW scheduler

The function `Spi_MainFunction_Handling()` should be called periodically only if polling mode is enabled for `Spi_AsyncTransmit()` .

### 6.2 Calls to notification functions, callbacks, callouts

#### Call-back Notifications:

None.

#### User Notification:

The SPI Handler & Driver provides notifications per job and sequence in asynchronous mode. The notifications can be configured as pointers to user defined functions. If notification is not desired, the appropriate `EndNotification` field shall be left blank.

For asynchronous transmissions, job and sequences notifications are performed before the scheduling of the next job (contrary to the recommendation given by SPI088) . In this way, calls like `Spi_SetupIB()` or `Spi_WriteIB()` can be targeted on the next schedulable jobs, before the starting of the job transfer.



## Chapter 7

# Memory Allocation

### 7.1 Sections to be defined in MemMap.h

#### For Config Data:

```
#ifndef SPI_START_SEC_CONFIG_DATA_<ALIGNMENT>
#define SPI_START_SEC_CONFIG_DATA_<ALIGNMENT>
#define MEMMAP_ERROR

/*Memory Section for Config Data to be defined here. Example given in the next line*/
#pragma ghs section rodata=".mcal_const_cfg"
#endif

#ifndef SPI_STOP_SEC_CONFIG_DATA_<ALIGNMENT>
#define SPI_STOP_SEC_CONFIG_DATA_<ALIGNMENT>
#define MEMMAP_ERROR

/*End of section to be mentioned here. Example given in the next line.*/
#pragma ghs section
#endif
```

Which the shortcut ‘<ALIGNMENT>’ means the variable alignment. In order to avoid memory gaps in the allocation variables are allocated according their size. Possible ALIGNMENT postfixes are described in the table at the end of this section.

#### For Code:

*In case of the code is in Flash:*

```
#ifdef SPI_START_SEC_CODE
#undef SPI_START_SEC_CODE
#undef MEMMAP_ERROR
/*Memory Section for Code to be defined here.*/
#endif
```

```
#ifdef SPI_STOP_SEC_CODE
#undef SPI_STOP_SEC_CODE
#undef MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
```

***In case of the code is in RAM instead of Flash memory:***

```
#ifdef SPI_START_SEC_RAMCODE
#undef SPI_START_SEC_RAMCODE
#undef MEMMAP_ERROR
/*Memory Section for Code to be defined here.*/
#endif
```

```
#ifdef SPI_STOP_SEC_RAMCODE
#undef SPI_STOP_SEC_RAMCODE
#undef MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
```

**For Variables:**

```
#ifdef SPI_START_SEC_VAR_< INIT_POLICY>_<ALIGNMENT>
```

```
#undef SPI_START_SEC_VAR_< INIT_POLICY>_<ALIGNMENT>
#undef MEMMAP_ERROR
/*Memory Section for Variables to be defined here.*/
#endif
```

```
#ifdef SPI_STOP_SEC_VAR_< INIT_POLICY>_<ALIGNMENT>
#undef SPI_STOP_SEC_VAR_< INIT_POLICY>_<ALIGNMENT>
#undef MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
```

Which the shortcut ‘<ALIGNMENT>’ means the same as used in config data section. The shortcut ‘<INIT\_POLICY>’ means the initialization policy of variables. Possible ‘<INIT\_POLICY>’ postfixes are described in the table at the end of this section.

#### **For Constant data:**

```
#ifdef SPI_START_SEC_CONST_<ALIGNMENT>
#undef SPI_START_SEC_CONST_<ALIGNMENT>
#undef MEMMAP_ERROR
/*Memory Section for Constants to be defined here.*/
#endif
```

```
#ifdef SPI_STOP_SEC_CONST_<ALIGNMENT>
#undef SPI_STOP_SEC_CONST_<ALIGNMENT>
#undef MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
```

Tables describe value range of shortcut **ALIGNMENT**, **INIT\_POLICY**:

**Table 7-1. Range of <ALIGNMENT>**

| <ALIGNMENT>        | Description                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BOOLEAN</b>     | Used for variables and constants of size 1 bit                                                                                                                                                                          |
| <b>8</b>           | Used for variables and constants which have to be aligned to 8 bit. For instance used for variables of size 8 bit or used for composite data types: arrays, structs and unions containing elements of maximum 8 bits    |
| <b>16</b>          | Used for variables and constants which have to be aligned to 16 bit. For instance used for variables of size 16 bit or used for composite data types: arrays, structs and unions containing elements of maximum 16 bits |
| <b>32</b>          | Used for variables and constants which have to be aligned to 32 bit. For instance used for variables of size 32 bit or used for composite data types: arrays, structs and unions containing elements of maximum 32 bits |
| <b>UNSPECIFIED</b> | Used for variables, constants, structure, array and unions when SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. For instance used for variables of unknown size                                           |

**Table 7-2. Range of <INIT\_POLICY>**

| <INIT_POLICY>  | Description                                                                            |
|----------------|----------------------------------------------------------------------------------------|
| <b>NO-INIT</b> | Used for variables that are never cleared and never initialized by start-up code (BSS) |
| <b>INIT</b>    | Used for variables that are initialized with values after every reset                  |

## 7.2 Linker command file

Memory shall be allocated for every section defined in MemMap.h.

## Chapter 8

# Configuration Parameter Considerations

Configuration parameter class for Autosar SPI driver fall into the following variants as defined below:

**Table 8-1. Configuration Parameters**

| Configuration Container  | Configuration Parameters | Configuration Variant    | Current Implementation |
|--------------------------|--------------------------|--------------------------|------------------------|
| SpiDriver                |                          |                          |                        |
|                          | SPI_MAX_CHANNEL          | PC, LT or PB             | Pre Compile (1)        |
|                          | SPI_MAX_JOB              | PC, LT or PB             | Pre Compile (1)        |
|                          | SPI_MAX_SEQUENCE         | PC, LT or PB             | Pre Compile (1)        |
| SpiChannel               |                          |                          |                        |
|                          | SpiChannelId             | Pre-Compile all Variants | Pre Compile            |
|                          | SpiChannelType           | PC, LT or PB             | Pre Compile (2)        |
|                          | SpiNbBuffers             | PC, LT or PB             | Pre Compile (3)        |
|                          | SpiDataWidth             | PC, LT or PB             | Post Build             |
|                          | SpiDefaultData           | PC, LT or PB             | Post Build             |
|                          | SpiEbMaxlength           | PC, LT or PB             | Post Build             |
|                          | SpiTransferStart         | PC, LT or PB             | Post Build             |
| SpiDemEventParameterRefs |                          |                          |                        |
|                          | Spi_E_Hardware_Error     | PC, LT or PB             | Post Build             |
| SpiExternalDevice        |                          |                          |                        |
|                          | SpiBaudRate              | PC, LT or PB             | Post Build             |
|                          | SpiCsIdentifier          | PC, LT or PB             | Post Build             |
|                          | SpiCsSelection           | PC, LT or PB             | Post Build             |
|                          | SpiCsPolarity            | PC, LT or PB             | Post Build             |
|                          | SpiDataShiftEdge         | PC, LT or PB             | Post Build             |
|                          | SpiEnableCs              | PC, LT or PB             | Post Build             |
|                          | SpiHwUnit                | PC, LT or PB             | Post Build             |
|                          | SpiShiftClockIdleLevel   | PC, LT or PB             | Post Build             |
|                          | SpiTimeClk2Cs            | PC, LT or PB             | Post Build             |
|                          | SpiTImeCs2Clk            | Vendor specific          | Post Build             |
|                          | SpiTimeCs2Cs             | Vendor specific          | Post Build             |

*Table continues on the next page...*

**Table 8-1. Configuration Parameters (continued)**

|               |                                  |                          |                 |
|---------------|----------------------------------|--------------------------|-----------------|
|               | SpiCsContinuous                  | Vendor specific          | Post Build      |
| SpiJob        |                                  |                          |                 |
|               | SpiJobId                         | Pre-Compile all Variants | Pre Compile     |
|               | SpiHwUnitSynchronous             | PC, LT or PB             | Post Build      |
|               | SpiJobEndNotification            | PC, LT or PB             | Post Build      |
|               | SpiJobStartNotification          | PC, LT or PB             | Post Build      |
|               | SpiJobPriority                   | PC, LT or PB             | Post Build      |
|               | SpiChannelAssignment             | PC, LT or PB             | Post Build      |
|               | SpiDeviceAssignment              | PC, LT or PB             | Post Build      |
| SpiSequence   |                                  |                          |                 |
|               | SpiSequenceId                    | Pre-Compile all Variants | Pre Compile     |
|               | SpiInterruptibleSequence         | PC, LT or PB             | Post Build      |
|               | SpiSeqEndNotification            | PC, LT or PB             | Post Build      |
|               | SpiJobAssignment                 | PC, LT or PB             | Post Build      |
| SpiGeneral    |                                  |                          |                 |
|               | SpiCancelApi                     | Pre-Compile all Variants | Pre Compile     |
|               | SpiChannelBuffersAllowed         | Pre-Compile all Variants | Pre Compile     |
|               | SpiDevErrorDetect                | Pre-Compile all Variants | Pre Compile     |
|               | SpiHwStatusApi                   | Pre-Compile all Variants | Pre Compile     |
|               | SpiInterruptibleSeqAllowed       | Pre-Compile all Variants | Pre Compile     |
|               | SpiLevelDelivered                | Pre-Compile all Variants | Pre Compile     |
|               | SpiVersionInfoApi                | Pre-Compile all Variants | Pre Compile     |
|               | SpiClockRef                      | Vendor specific          | Pre Compile (4) |
|               | SpiGlobalDmaEnable               | Vendor specific          | Pre Compile     |
|               | SpiSyncTransmitTimeout           | Vendor specific          | Pre Compile     |
|               | SpiOptimizeOneJobSequences       | Vendor specific          | Pre Compile     |
|               | SpiOptimizedSeqNumber            | Vendor specific          | Pre Compile     |
|               | SpiOptimizedChannelsNumber       | Vendor specific          | Pre Compile     |
|               | SpiSupportConcurrentSyncTransmit | Vendor specific          | Pre Compile     |
| SpiNonAUTOSAR |                                  |                          |                 |
|               | SpiEnableHWUnitAsyncMode         | Vendor specific          | Pre Compile     |
|               | SpiEnableDualClockMode           | Vendor specific          | Pre Compile     |
|               | SpiAlternateClockRef             | Vendor specific          | Pre Compile     |
|               | SpiAllowBigSizeCollections       | Vendor specific          | Pre Compile     |
|               | SpiForceDataType                 | Vendor specific          | Pre Compile     |
|               | SpiDisableDemReportErrorStatus   | Vendor specific          | Pre Compile     |
| SpiPhyUnit    |                                  |                          |                 |
|               | SpiPhyUnitMapping                | Vendor specific          | Pre Compile     |

Table continues on the next page...

**Table 8-1. Configuration Parameters (continued)**

|  |                       |                 |            |
|--|-----------------------|-----------------|------------|
|  | SpiPhyUnitSync        | Vendor specific | Post Build |
|  | SpiPhyUnitMode        | Vendor specific | Post Build |
|  | SpiPhyUnitAsyncMethod | Vendor specific | Post Build |
|  | SpiPhyTxDmaChannel    | Vendor specific | Post Build |
|  | SpiPhyTxDmaChannelAux | Vendor specific | Post Build |
|  | SpiPhyRxDmaChannel    | Vendor specific | Post Build |

1. Adding or removing Channels, Jobs or Sequences typically requires updating the application, rendering those parameters useless as PB option.
2. Changing the buffer type of a channel requires updating the application, rendering this parameter useless as PB option.
3. Changing the size for internal buffers post build requires a “PostBuild RAM” concept.
4. Please note that this is the peripheral clock frequency supplied to the DSPI.



## Chapter 9

# Integration Steps

This section gives a brief overview of the steps needed for integrating SPI:

1. Generate the required SPI configurations. For more details refer to the section "[Setting up the Plug-ins](#)"
2. Allocate proper memory sections in MemMap.h and linker command file. For more details refer to the section "[Memory Allocation](#)"
3. Make sure all include files for compilation are as per the section "[Files required for Compilation](#)"
4. Map the ISRs to their vector locations. For more details refer to the section "[ISR to configure within OS – dependencies](#)"
5. Compile & build the SPI with all the dependent modules. For more details refer to the sections "[Building the Driver](#)" & "[ISR to configure within OS – dependencies](#)"

**Note:**MCU shall be initialized with desired global Pre-scalar and system frequency before initializing the SPI driver. PORT shall be initialized with desired signal settings for DSPI.



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